ANALYST

Haowen, Maya, Dina

The problem: correct score, wrong patch image

Independent sorting could break the link between a patch embedding and the PNG displayed.

ACTUAL EMBEDDING ORDER		INDEPENDENTLY RE-SORTED PNGS	
00	patch_1.png	00	patch_1.png
01	**patch_10.png**	01	**patch_2.png**
02	patch_2.png	02	patch_10.png

Same index ≠ same source patch

Root causes: filenames were dropped during loading; plots reconstructed image order separately; missing features could be silently skipped.

The fix: preserve patch identity end to end

Each feature vector now retains its exact PNG path through the attention calculation and plots.

01

dataset.py

Load feature + store original path

`vecs[i] ↔ loaded_patch_paths[i]`

→

02

attention_analysis.py

Attach path to each attention record

`effective score + patch_path`

→

03

Case / slice plots

Read the stored PNG path directly

`no independent re-sorting`

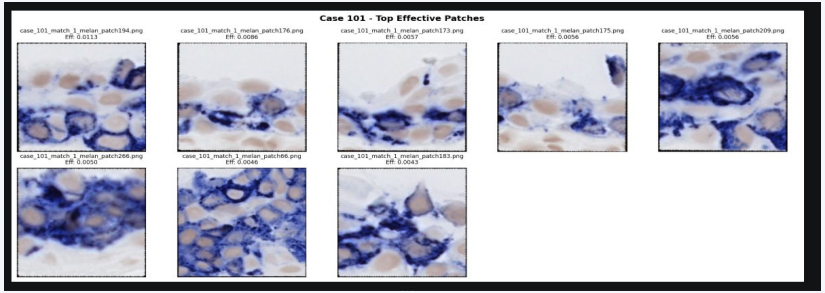
Effective patch attention = stain weight × slice weight × patch weight

Attention formula unchanged. The improvement is correct filename-to-score alignment.

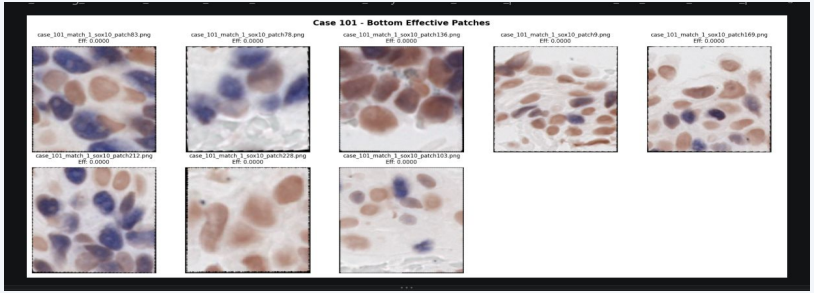
Earlier checkpoint validation: plots now match filenames

October 28 checkpoint run • 17 test cases • corrected case-level top / bottom visualization.

TOP EFFECTIVE ATTENTION · CASE 101



BOTTOM EFFECTIVE ATTENTION · CASE 101



17 / 17
cases evaluated

88.24%
classification accuracy

166
attention-analysis PNGs

Earlier successful run used an exclusion-based missing-embedding workflow; this is separate from the new strict-audit runs.

New component: selected-patch embedding audit

After the professor clarified that small patches may lack embeddings, we audit what the loader actually selects.

1

Apply selection

Use the test split, slice limit, and per-slice cap

→

2

Check selected PNGs

Look up the expected .pt feature path for each

→

3

Report + stop

Write all missing paths to CSV, then raise one summary error

The scan continues past missing files; the model evaluation stops only after the complete preflight audit.

Scope: deterministic test selection. Training shuffling requires a separate selection-specific audit.

Latest Quest audit: 370 selected embeddings missing

November 5 • 18 test cases • complete selected-patch preflight scan • job 7344694.

15,519	370	97.62%
selected patches checked	missing .pt features	filename coverage

SELECTED PATCHES



15,149 available

370 missing (2.38%)

CSV: preflight_missing_embeddings_test.csv • Includes case, stain, slice, PNG path, expected .pt path

Validation: tests, source files, and remaining limitation

Each check has a dedicated test file; full inference still stops under the strict missing-embedding policy.

PASS	Patch ↔ embedding	test_patch_mapping.py
PASS	Attention ↔ filename	test_attention_mapping.py
PASS	Selected-patch detection	test_targeted_audit.py
PASS	Multi-worker CSV logging	test_audit_dataloader.py
PASS	Quest preflight audit	audited_patch_filter.py + main.py

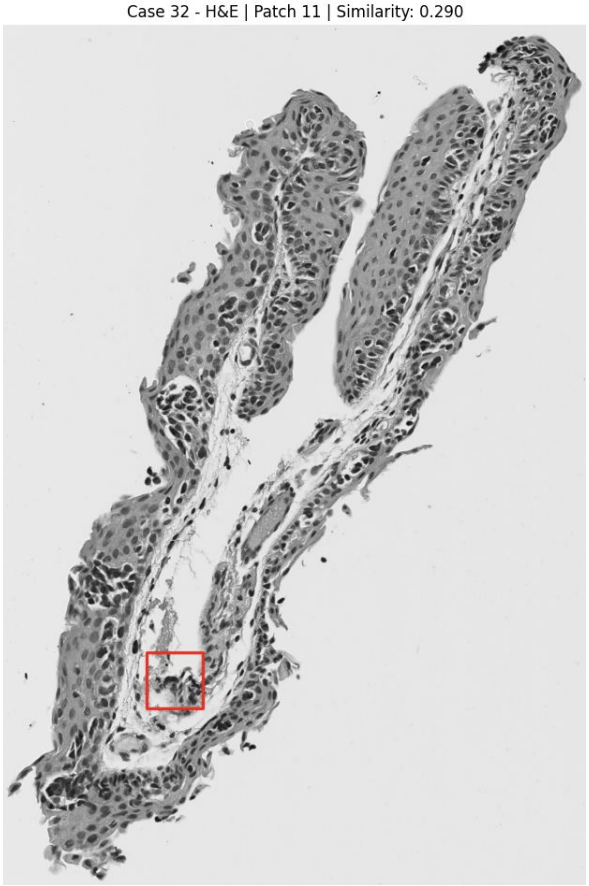
https://github.com/arvindkrishna87/Team4_TheAnalysts/tree/main/Presentation2/1_Presentation2_base_code_2026_11_4_Haowen_Tan

Code Reference: https://docs.google.com/document/d/1b7RU5zjXpkFcOOsq0wZw9HkZlADDiTcZ/edit

Takeaway: identity tracking and targeted audit validated; latest full evaluation intentionally stops on missing .pt files.

Maya Schwartz

Showing Patch Location on Slice



Initial approach: OpenCV template matching (low similarity score, unreliable results)

2 NEW .PY FILES

generate_patch_coordinates.py → outputs CSV

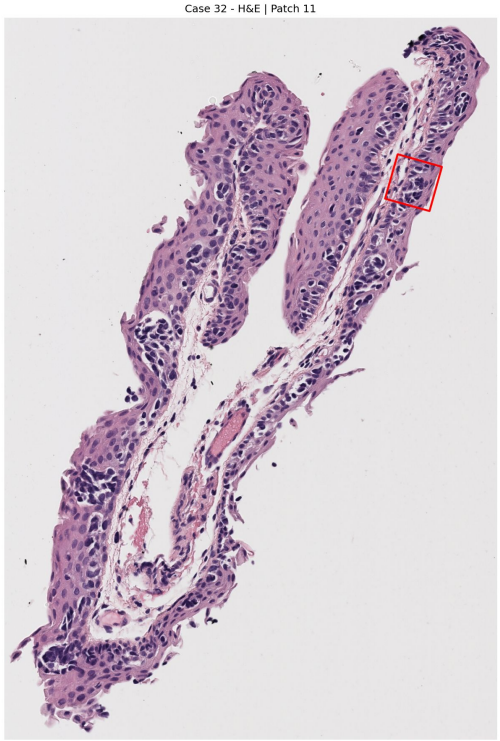
- Input: Original slice annotation + slice image
- Uses original patch-placement algorithm to save patch corner coordinates

visualize_slice_patch_index.py → outputs annotated image

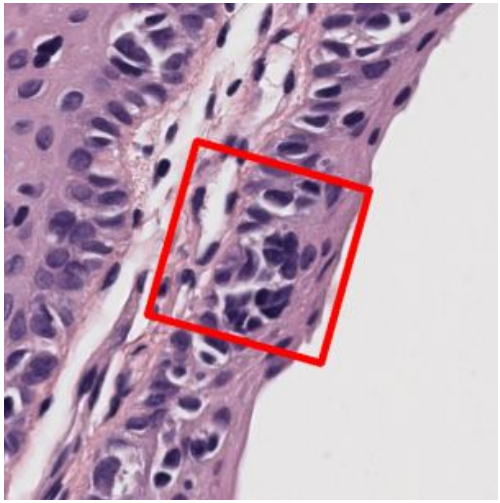
- Locates selected patch by index
- Outlines patch on full slice and saves image

Maya Schwartz

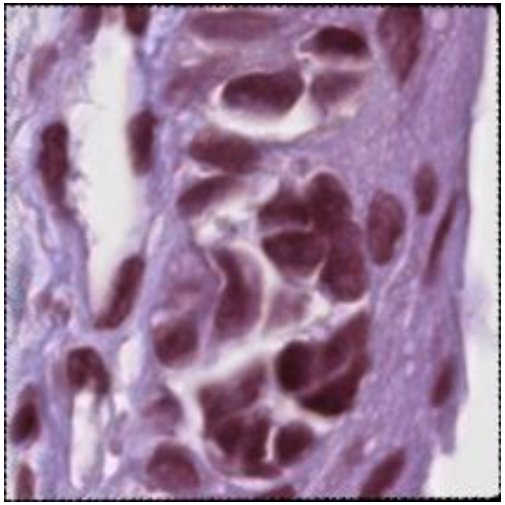
Case 32 Match 1 h&e Patch 11



generated image
(full tissue slice)



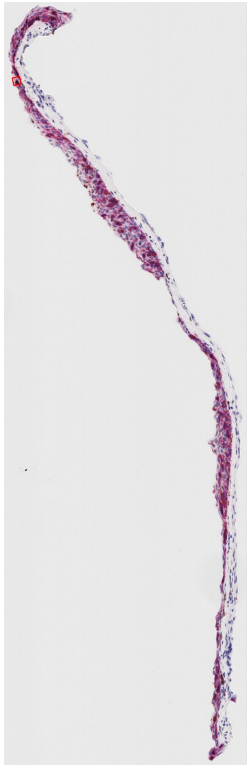
generated image
(zoomed in)



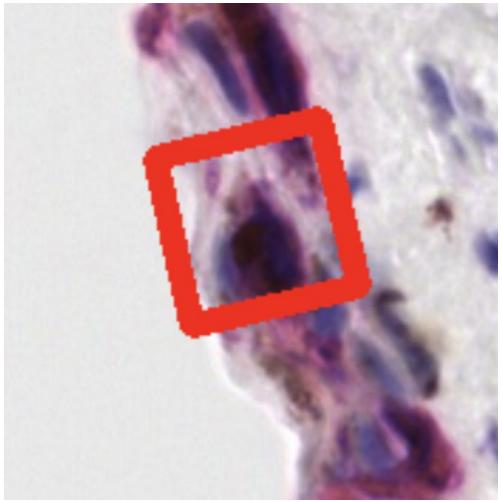
patch 11
(for comparison)

Maya Schwartz

Case 5 Match 1 Melan Patch 12



generated image
(full tissue slice)



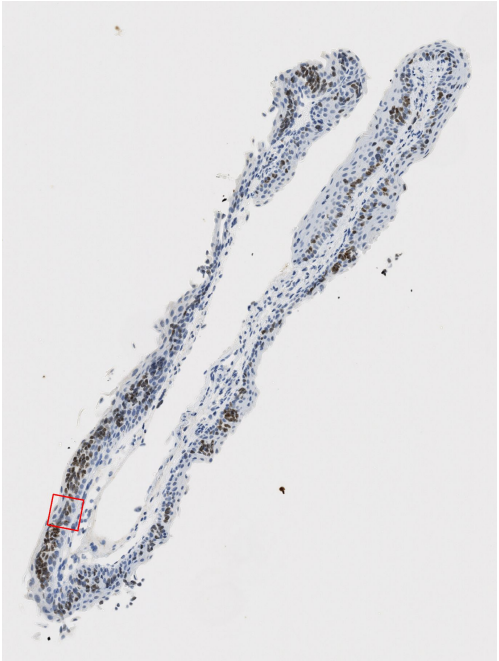
generated image
(zoomed in)



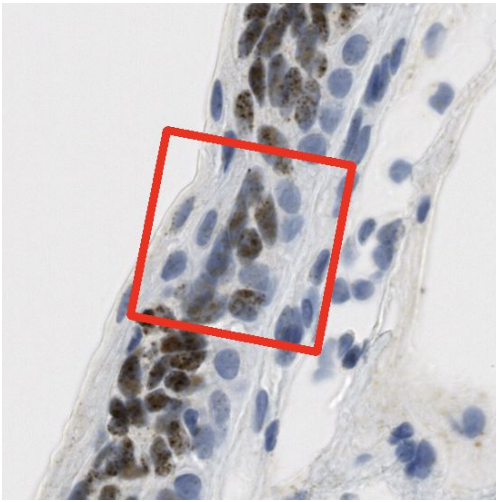
patch 12
(for comparison)

Case 32 Match 1 sox10 Patch 33

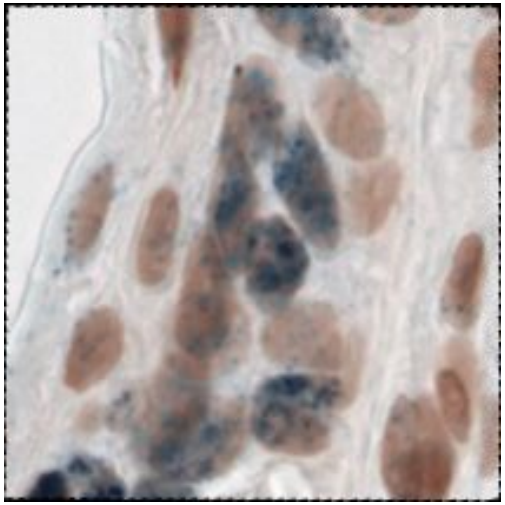
Maya Schwartz



generated image
(full tissue slice)



generated image
(zoomed in)



patch 33
(for comparison)

Next Steps

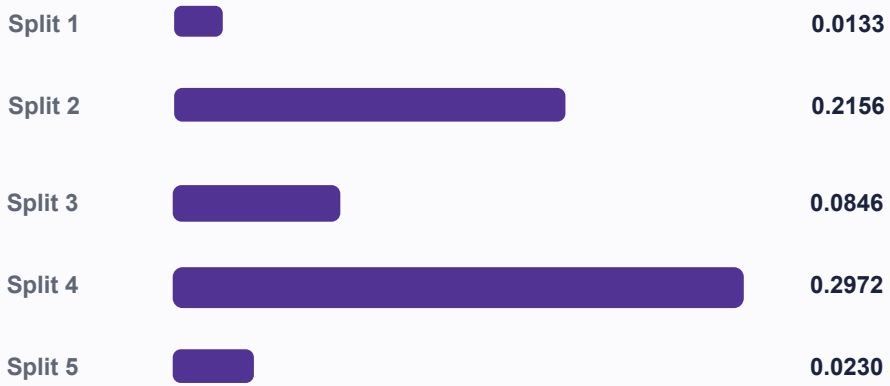
- **Visualize** high attention selected patches to identify potential patterns
- Investigate **unreadable** TIFF files (e.g., case 36), potentially caused by file corruption or unsupported formatting (TIFF decoding issue)
- **Storage** of csv files when using with a large number of slices

Choice 3: Splits that do not memorize

Why do some splits resist overfitting?

Team 2: contrasting validation-loss behavior across five historical splits

Best recorded validation loss by split



The anomaly

Split 5 reached 0.0230 at epoch 15 and remained relatively low afterward

Important nuance

Split 1 also reached a very low minimum (0.0133)

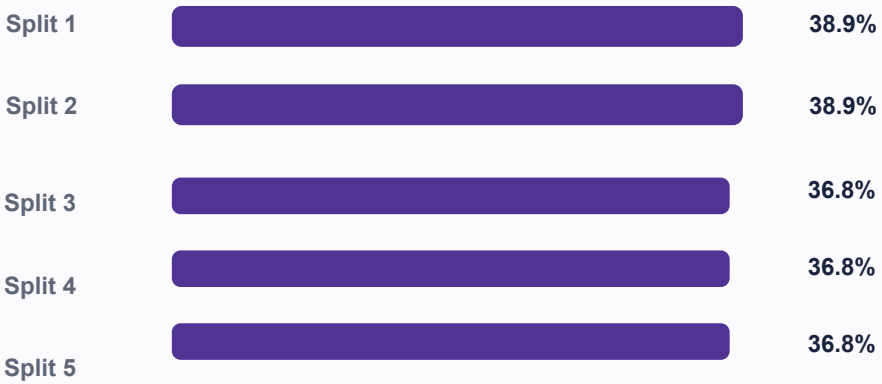
Question: Can class composition or tissue-data quantity account for the difference?

Two explanations tested

Descriptive comparisons using Team 2 original 93-case split files

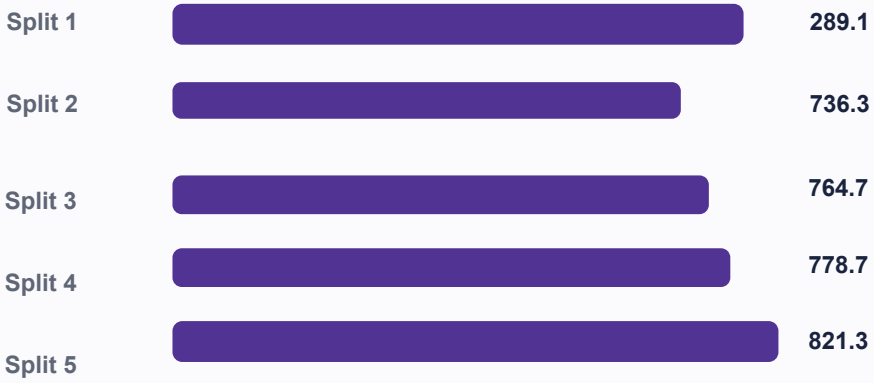
A Class composition

Validation set: % benign



B Training patch quantity

Average capped patch positions per case



Conclusion

Class balance is unlikely to explain the pattern; patch counts are inconclusive.

Patch counts reconstructed from currently available PNGs with 500-patch / 5-slice caps

From aggregate loss to case-level evidence

What we found in the training code, and the next analysis to run

01 Limitation Found

Validation calculates per-case loss but saves only epoch averages

02 Checkpoint located

Saved Split 5 best.pth is epoch 15; recorded validation loss = 0.0230188

03 Next steps

Re-evaluate original validation cases with saved models; compare per-case losses across splits

Takeaway

No definitive explanation yet. We ruled out simple class-balance differences, found a patch-count signal worth testing, and established a reproducible next step.

Consultant.1

Hortense Ineza Mutamba

Model Input Size Analysis

“Does UNI2’s input size matter?”

01 / Question

How do UNI2’s training image size (224×224) and its handling of variable-sized images affect performance on our variable-sized epithelial patches?

02 / Answer & Key Finding

UNI loses much less accuracy than older models when image size changes, but this is only shown on large images.

Research Gap: Nobody has shown it for tiny patches like some of ours.

UNI2 Architecture & Mechanism

How UNI2 Turns Any Image Into One Embedding

01 / Input Sizing

Flexible Image Dimensions

The image is resized or kept at its original native size.

02 / Grid Patching

14×14 Pixel Squares

Cut into a grid of 14×14 px squares.

- 224 px image → 16×16 = 256 squares
- 112 px patch → 8×8 = 64 squares

03 / Encoding

Vectors & Position Tags

Each square becomes a number list, and a position tag records where it sat in the grid.

04 / Attention

Transformer & CLS Token

Squares look at each other via self-attention. An extra summary slot (**CLS token**) collects the overall picture.

05 / Output

Fixed Embedding Size

The CLS output is the final embedding: always **1,536 numbers**, regardless of input image size.

Key Takeaway

More Squares, Same Embedding

A bigger image yields more squares, not a bigger embedding.

`dynamic_img_size`

lets position tags stretch to fit.

Source: Mahmood Lab (n.d.), UNI2-h model card.

Model Comparison

Key Differences: KimiaNet vs. UNI2

01 / Architecture

CNN vs. Transformer

KimiaNet: Shrinks image 32× (tiny patch becomes ~1 feature "cell").

UNI2: Cuts the image into 14×14-pixel squares

02 / Input Dimensions

1000×1000 vs. 224×224

KimiaNet: Designed for large 1000×1000 inputs.

UNI2: Standardized at 224×224 (default pipeline resizes inputs to fit).

03 / Feature Output

4,096 vs. 1,536 Embeddings

KimiaNet: 4,096 numbers (projector shifts 4,096 → 512).

UNI2: 1,536 numbers (projector shifts 1,536 → 512).

04 / Training Domain

TCGA Slides vs. 200M+ Images

KimiaNet: Trained specifically on TCGA pathology slides.

UNI2: Pre-trained on 200M+ tiles from 350k+ slides (model card)

Patch Impact (So For Us): at native size UNI2 sees 32 px → 2×2 squares, 64 px → 4×4, 128 px → 9×9, typical 199 px → 14×14, and 224 px → 16×16.

Sources: Mahmood Lab (n.d.); Riasatian et al. (2021).

Literature Review

What the Published Evidence Says

01 / Key Finding

UNI held up robustly across 224 to 1,792 px (Chen et al. 2024, p. 856, Fig. 3d–e). Accuracy drops significantly less than competing models when resolution changes.

BRCA subtyping drop when images shrink from 1,344 px to 224 px.

UNI-6.3%

CTransPath-18.8%

REMEDIS-32.5%

02 / Author Warning & Limits

Warning: Resizing images directly to 224 px alters the effective magnification (mpp) of the tissue.

Current Study Limitations

- **Large Images Only:** Tested exclusively on images starting from 224 px and up.
- **No Small Patch Upscaling:** Nothing smaller than 224 px was upscaled in published benchmarks.
- **Model Version:** UNI2-h specifically was not tested in these findings.

Strategic Implications

What This Means for Our Data & Recommendations

01 / Data Audit Context

23.3%

of 81,627 patches are under 128 px (patch-size audit, next slides).

02 / Core Resizing Dilemma

Upscaling small patches (< 128 px): Adds no real structural detail to UNI2.

Downscaling large patches: Loses high-resolution detail. Both drastically alter effective tissue scale (mpp).

(Rea, 2022)

03 / Actionable Recommendations

- *Keep real patch size using UNI2's dynamic-size mode, instead of forcing 224×224.*
- *Raise the minimum patch size from 32 px to 128 px (keeps 76.7% of patches).*
- *Confirm slide magnification: UNI was pretrained at ×20 (Chen et al., 2024).*

04 / Honest Research Limit

No published benchmarks directly evaluate performance on patches of our exact size distribution.

Next Step:
Our upcoming **UNI2 vs. KimiaNet empirical experiment** is required to definitively settle optimal preprocessing.

Documentation

References

1. Chen, R. J., Ding, T., Lu, M. Y., Williamson, D. F. K., Jaume, G., Song, A. H., Chen, B., Zhang, A., Shao, D., Shaban, M., Williams, M., Oldenburg, L., Weishaupt, L. L., Wang, J. J., Vaidya, A., Le, L. P., Gerber, G., Sahai, S., Williams, W., & Mahmood, F. (2024). Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30(3), 850–862. https://doi.org/10.1038/s41591-024-02857-3
2. Mahmood Lab. (n.d.). *UNI2-h* [Model card]. Hugging Face. https://huggingface.co/MahmoodLab/UNI2-h
3. Mahmood Lab. (n.d.). *UNI: Towards a general-purpose foundation model for computational pathology* [Source code repository]. GitHub. https://github.com/mahmoodlab/UNI
4. Riasatian, A., Babaie, M., Maleki, D., Kalra, S., Valipour, M., Hemati, S., Zaveri, M., Safarpoor, A., Shafiei, S., Afshari, M., Rasoolijaberi, M., Sikaroudi, M., Adnan, M., Shah, S., Choi, C., Damaskinos, S., Campbell, C. J. V., Diamandis, P., Pantanowitz, L., Kashani, H., Ghodsi, A., & Tizhoosh, H. R. (2021). Fine-tuning and training of DenseNet for histopathology image representation using TCGA diagnostic slides. *Medical Image Analysis*, 70, 102032.
5. Rea, K. (2022, October 12). *How to select the correct magnification and patch size for digital pathology projects*. Andrew Janowczyk. https://andrewjanowczyk.com/how-to-select-the-correct-magnification-and-patch-size-for-digital-pathology-projects/

Consultant.2

Alzaharaa Alzaabi

Zooming In: Patches Entering UNI2

Slides + QuPath
Epithelium outlined

→

Variable-size patches
Kept if 32 px or more

→

Resize to 224 px?
Step in UNI2's guide

→

UNI2
Cuts image into 14 px squares

→

Attention MIL
1,536 numbers per patch

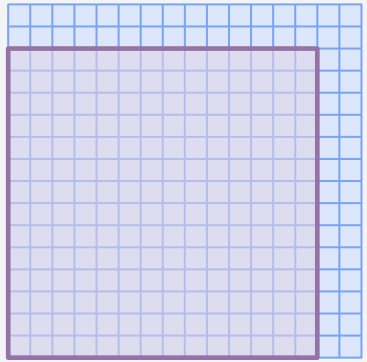
01 **199 px vs 224 px:** our typical patch is close to UNI2's size

02 **14 px squares:** UNI2 cuts each image up, then sums it as 1,536 numbers

03 **Resize step:** UNI2's guide forces every patch to 224 px

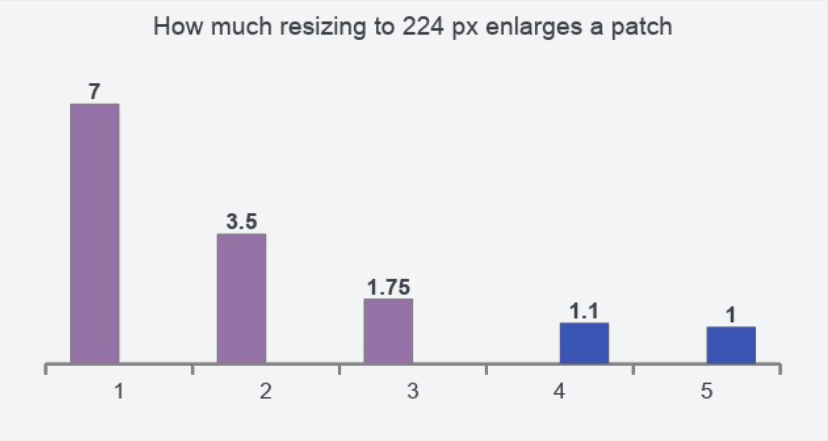
04 **32 px → 7×:** our smallest patches get stretched 7 times

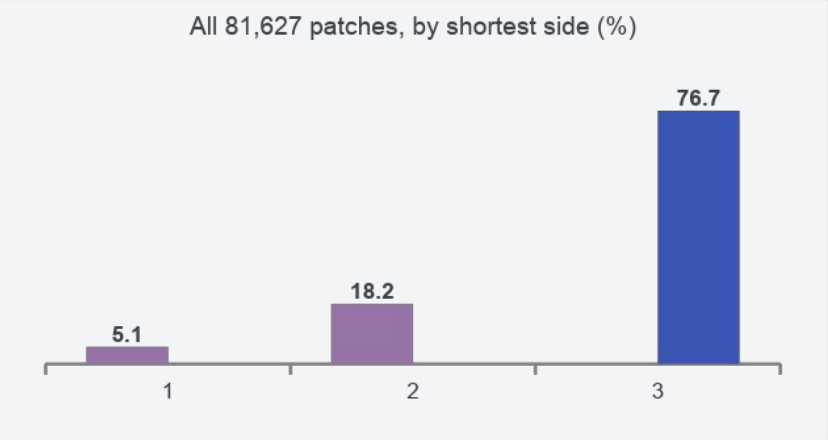
Drawn to scale: squares UNI2 sees



Typical patch
199 px = 14 × 14

Smallest patch
32 px = 2 × 2





Evidence

How Much Would Patches Stretch?

01 **7×** how much a 32 px patch is blown up

02 **3.5×** for a 64 px patch

03 **23.3%** of patches stretched 1.75× or more

04 **1.1×** for our typical patch: barely changes

What resizing would do to our patches



- < 64 px: stretched 3.5×+ (5.1%)
- 64–127 px: stretched 1.75×+ (18.2%)
- 128 px +: small change (76.7%)

Input Size: Weakness, Impact, and Fix

The weakness

224 px

UNI2's guide resizes every patch to this size. Our code doesn't do this today.

Why it matters

7×

A 32 px patch gets blown up 7 times. No new detail; the cells just look huge.

What research says

6.3%

UNI's drop when images were shrunk to 224 px, vs 18.8% and 32.5% for older models. Never tested on stretching.

Chen et al., 2024

The fix

128 px

Keep real size and raise the minimum to 128 px (line 32). Keeps 76.7% of patches.

How to Test the Fix

01 **Run it two ways**

Option A resizes to 224 px. Option B keeps real size. Same seed, same split.

02 **Compare the same patches**

Large patches should match both ways. Watch whether the small ones change.

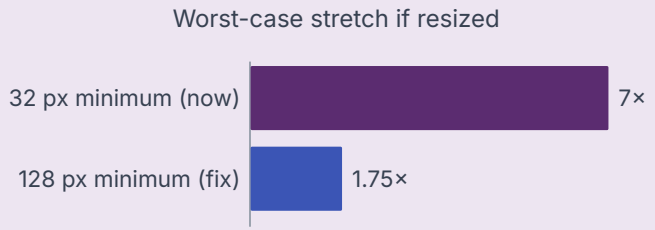
03 **Check accuracy and recall**

Per class. Keep whichever option scores better.

04 **Look at top attention patches**

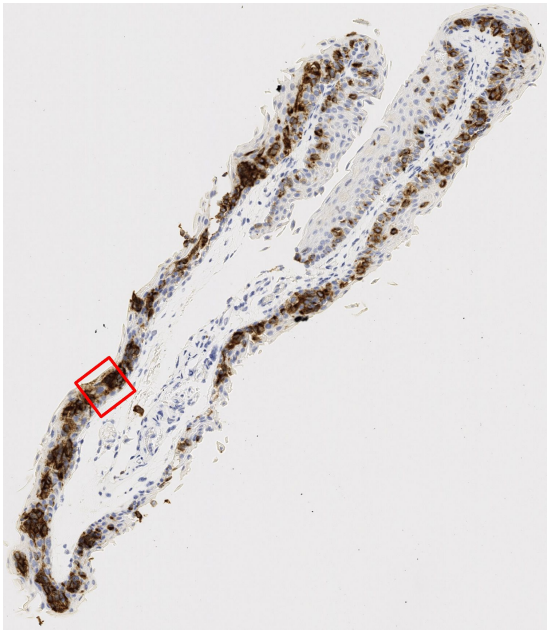
The top 8 per case should show real cells, not blurry blow-ups.

If small patches come out different: size matters for UNI2, so keep real size and the 128 px minimum. Also confirm our slides were scanned at 20×, the zoom UNI2 learned from.

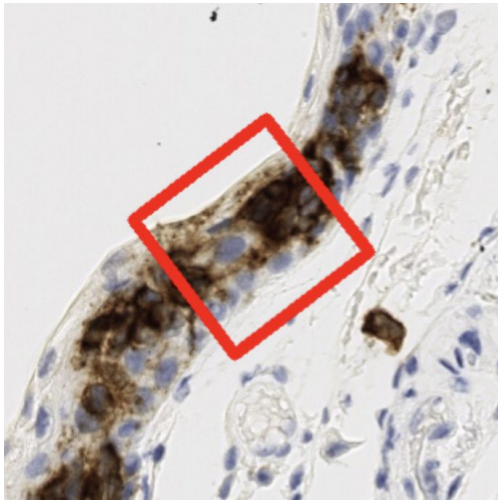


Appendix

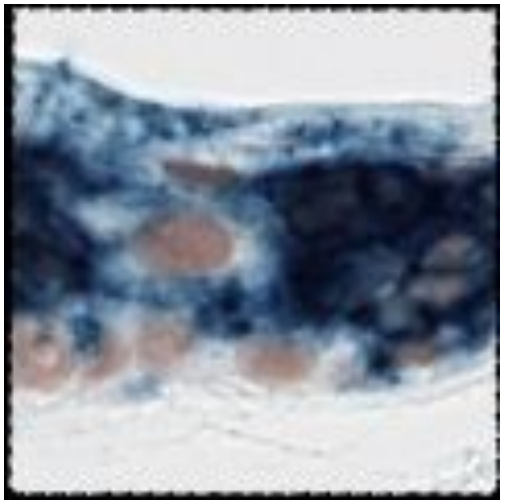
Case 32 Match 1 Melan Patch 45



generated image
(full tissue slice)



generated image
(zoomed in)



patch 11
(for comparison)

Case 5 Match 1 h&e Patch 16



generated image
(full tissue slice)



generated image
(zoomed in)



patch 16
(for comparison)

THANK YOU

Presenter

TEAM 4

Q & A

Open for Questions & Discussion